MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, GERMANIUM, SWITCHING

TYPE 2N167A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a NPN, germanium, low-power switching transistor.

1.2 Physical dimensions. See figure 1

1.3 Maximum ratings.

P_T 1/	V_{CEO}	V_{EBO}	I_C	I_E	T_A
mW	Vdc	Vdc	mAdc	mAdc	°C
70	27	5	75	-75	-65 to +85

1/ Derate linearly 1.167 mW/°C for $T_A > 25$°C.

1.4 Primary electrical characteristics.

Limits	C_{obo} V_{CB} = 5 Vdc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	f_{hfb} V_{CB} = 5 Vdc, I_C = 1 mAdc	h_{FE} V_{CE} = 1 Vdc, I_C = 5 mAdc	t_{on}	t_{off}
	pF	MHz		ns	ns
Min	---	5	17	---	---
Max	6	25	90	60	220

2. APPLICABLE DOCUMENTS

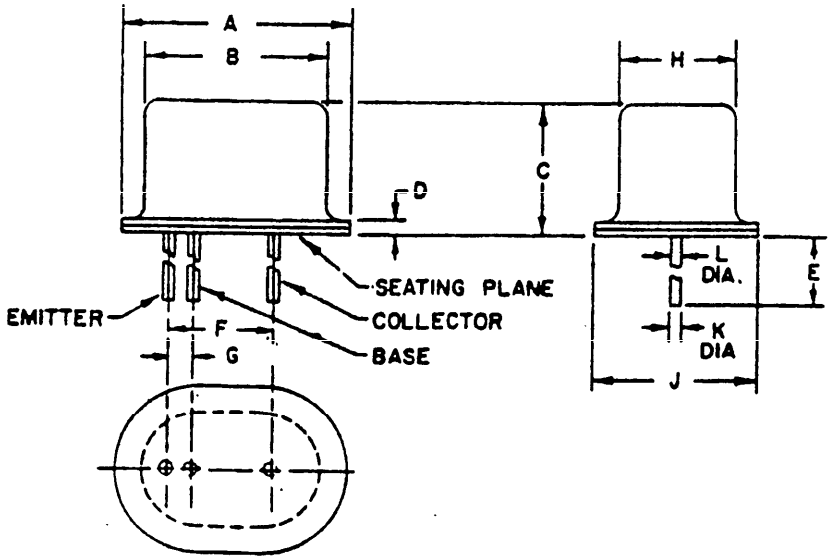
2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

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DIMENSIONS					
	INCHES		MILLIMETERS		
LTR	MIN	MAX	MIN	MAX	NOTES
A	.395	.530	10.03	13.46	
B	.335	.375	8.51	9.53	
C	.300	.320	7.62	8.13	
D	---	.036	---	.91	
E	1.500	1.750	38.10	44.45	4, 5
F	.172	.212	4.37	5.38	
G	.038	.058	.97	1.47	
H	.195	.235	4.95	5.97	
J	.255	.395	6.48	10.03	
K	.016	.021	0.41	0.53	2, 5
L	.016	.019	0.41	0.48	3, 5

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. All leads electrically insulated from the case.
5. All three leads.

FIGURE 1. Physical dimensions of transistor type 2N167A

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FIGURE 2. Physical dimensions surface mount devices.

STANDARD

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MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

V_{BA} - - - Channeling voltage.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hours life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria (see 4.3.3).

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4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

4.4.1 Channeling voltage test. The circuit of figure 2 is used to insure absence of channeling when the transistor is operated at low power levels.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Breakdown voltage, collector to emitter	3011	Bias cond. D; I_C = 300 µAdc		BV_{CEO}	27	---	Vdc
Breakdown voltage, emitter to base	3026	Bias cond. D; I_E = 1.5 µAdc		BV_{EBO}	5	---	Vdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 15 Vdc		I_{CBO}	---	2	µAdc
Subgroup 3			5				
Forward-current transfer ratio	3076	V_{CE} = 1 Vdc; I_C = 5 mAdc		h_{FE}	17	90	---
Forward-current transfer ratio	3076	V_{CE} = 1 Vdc; I_C = 0.5 mAdc		h_{FE}	25	---	---
Small-signal short-circuit forward-current transfer ratio cutoff frequency	3301	V_{CB} = 5 Vdc; I_C = 1 mAdc		f_{hfb}	5	25	MHz
Open circuit output capacitance	3236	V_{CB} = 5 Vdc; I_E = 0; 100 kHz≤f≤1 MHz		C_{obo}	---	6	pF
Channeling voltage	---	(See figure 2 and 4.4.1)		V_{BA}	---	0.4	Vdc
Subgroup 4			5				
Pulse response:							
Turn-on time	3251	Test cond. B (see figure 3)		t_{on}	---	60	ns
Turn-off time	3251	Test cond. B (see figure 3)		t_{off}	---	220	ns
Subgroup 5			10				
High-temperature operation:		T_A = 71°C					
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 15 Vdc		I_{CBO}	---	29	µAdc
Low-temperature operation:		T_A = -55°C					
Forward-current transfer ratio	3076	V_{CE} = 1 Vdc; I_C = 5 mAdc		h_{FE}	10	---	---

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TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			20				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. A, except in step 1, T_A = -65°C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A, except T_{max} = 85°C $^{+0}_{-5}$°C		---	---	---	---
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 15 Vdc		I_{CBO}	---	2	µAdc
Forward-current transfer ratio	3076	V_{CE} = 1 Vdc, I_C = 5 mAdc		h_{FE}	17	90	---
Subgroup 3			20				
Shock	2016	Nonoperating; 500 G; 1 ms, 1 blow in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration, variable frequency	2056	10 G		---	---	---	---
Constant acceleration	2006	10,000 G in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
End points:							
Hermetic seal	1071	Test cond. G or H for fine leaks; test cond. A, C, D or F for gross leaks		---	---	$1x10^{-7}$	atm cc/sec
Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 6			10				
High-temperature life (nonoperating)	1032	T_{stg} = +85°C; time = 340 hours (see 4.3.4)		---	---	---	---

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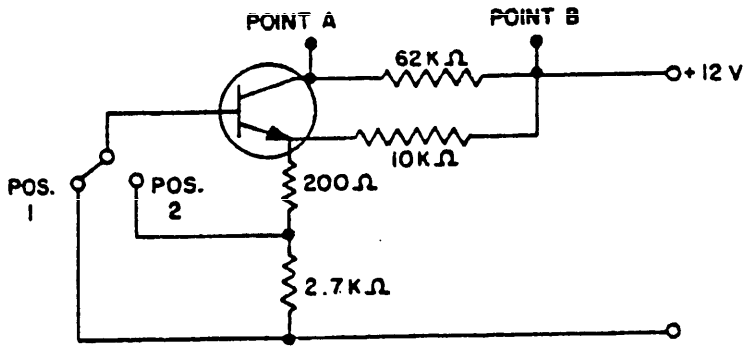
TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 6 - Continued							
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 15 Vdc		I_{CBO}	---	3	μAdc
Forward-current transfer ratio	3076	V_{CE} = 1 Vdc; I_C = 5 mAdc		h_{FE}	10	---	---
Subgroup 7			10				
Intermittent operation life	1037	V_{CB} = 10 Vdc (min); I_E = 7 mAdc; Cycle: "on" 50 ± 2 minutes "off" 10 ± 2 minutes time = 340 hours (see 4.3.4)		---	---	---	---
End points:							
(Same as subgroup 6)							

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +85°C (see 4.3.4)		---	---	---	---
End points:							
(Same as subgroup 6 of group B)							
Subgroup 2			λ = 10				
Intermittent operation life	1036	V_{CB} = 10 Vdc (min); I_E = 7 mAdc; Cycle: "on" 50 ± 2 minutes "off" 10 ± 2 minutes		---	---	---	---
End points:							
(Same as subgroup 6 of group B)							

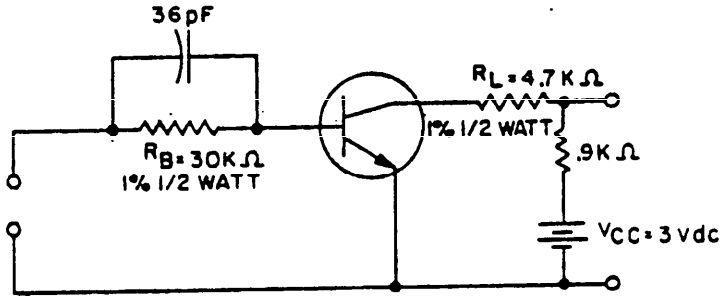
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TEST PROCEDURE:

(a) With the switch in position 1 and 12V applied to the circuit, the transistors shall be left operating for approximately 24 hours at room temperature.
(b) After the 24-hour period of operation, switch to position 2.
(c) Measure the voltage from point B to point A.

FIGURE 2. Channeling voltage test circuit.



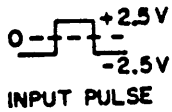


FIGURE 3. Pulse response circuit.

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5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MU, MI
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - SM
Navy - CG, MC, OS, AS
Air Force - 13, 15, 19, 70, 80

Preparing activity:
Navy - EC

Agent:
DSA - ES

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